Deep Learning-Based Road Damage Detection and Classification for Automated Road Condition Monitoring

Synopsis report on
Deep Learning Project
[ICT-4442]

Submitted By

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August 2026

Brief Description of the Project:

Road networks deteriorate continuously due to traffic loading, weathering, and material fatigue, manifesting as cracks, potholes, ravelling, and patches. Manual inspection by municipal survey teams is slow, subjective, costly, and does not scale to large road networks, leading to delayed maintenance and increased accident risk. This project develops and compares multiple deep learning architectures for automated detection, localization, and classification of road surface damage from both still images and video (dashcam/drone frame sequences), enabling continuous, low-cost road condition monitoring for smart-city and highway-authority applications.

The project uses a publicly available multi-national road damage image dataset supplemented with video frame extraction for temporal coverage. We will implement and compare 3-4 deep learning models spanning distinct architecture families - a CNN-based classification baseline, a one-stage object detector, an encoder-decoder segmentation network, and an attention/transformer-based model - each assigned to a different team member. All models will be evaluated under a common train/validation/test protocol using standard metrics (accuracy, precision, recall, F1-score, mAP, and IoU as applicable), with a final comparison table identifying which architecture family is best suited for real-time, deployable road condition monitoring.

Dataset(s) Identified (with details of the dataset):

Primary dataset: RDD2022 (Road Damage Dataset 2022 / CRDDC'2022), a publicly available multi-national image dataset containing 47,420 road images collected from six countries (India, Japan, Czech Republic, Norway, United States, China), annotated with bounding boxes for four damage classes - Longitudinal Crack (D00), Transverse Crack (D10), Alligator Crack (D20), and Pothole (D40). Images are in .jpg format with PASCAL-VOC style XML annotations, suitable for both detection (YOLO) and, after mask conversion, segmentation (U-Net) tasks.

Video component: short dashcam / drone video clips (self-collected or from open road-survey footage) will be decomposed into frames at a fixed sampling rate to extend the image pipeline to video-based monitoring and to test model robustness on frame sequences rather than curated still photographs.

The dataset will be split into train/validation/test (e.g., 70/20/10) with the same split reused across all models to ensure a fair comparison.

Proposed Models for Comparison:

1. CNN Classification Baseline - ResNet-50 / MobileNetV2 (transfer learning): classifies image patches as damaged / undamaged and by damage type; represents the classical CNN family.

2. One-Stage Object Detector - YOLOv8: detects and localizes damage with bounding boxes directly on full images/video frames; represents the detection-architecture family.

3. Encoder-Decoder Segmentation Network - U-Net (with a ResNet encoder): produces pixel-level crack/pothole masks for area and severity estimation; represents the segmentation-architecture family.

4. Attention/Transformer-Based Model - Vision Transformer (ViT) or a Transformer-U-Net hybrid (e.g., TransUNet): represents the attention-based family, since a classical RNN/LSTM is not a natural fit for spatial image data.

This spans four distinct architecture families as required (classification CNN, one-stage detector, encoder-decoder segmentation, attention-based), giving each member an independently demonstrable model.

Model-to-Member Assignment (one model/task per member):

Lakshya Jain: ResNet-50 / MobileNetV2 classification baseline.
Dhruv Maheshwari: YOLOv8 detection model.
Kumar Kshitij: U-Net segmentation model.
MANAS LAXMAN KUDIPADY: Vision Transformer / TransUNet attention-based model.

Existing State-of-the-Art Method(s) Related to the Proposed Project:

Prior work on the RDD2022/RDD2020 benchmark shows YOLO-family detectors dominate real-time road damage detection: YOLOv7-based ensembles with coordinate attention reported F1 scores above 80% on U.S. Street-View data, while lightweight YOLOv8 variants (YOLOv8-PD, RDD-YOLO, YOLO-RD) use attention modules (SimAM, LSKA) and lightweight convolutions (GhostConv) to push mAP@0.5 above 70% with under 3M parameters, making them suitable for edge deployment. For pixel-level analysis, U-Net-based segmentation models with ResNet encoders (RUC-Net) and attention-augmented skip connections achieve strong F1/IoU scores on crack datasets such as CFD, Crack500 and DeepCrack. More recent work combines transformer blocks with U-Net decoders (e.g., TransUNet-style hybrids) to capture long-range context in crack patterns, motivating our inclusion of an attention-based model in the comparison.

Objectives:

1. To design and implement 3-4 deep learning architectures spanning distinct families - CNN classification, one-stage detection, encoder-decoder segmentation, and attention/transformer-based - for identifying and localizing road surface damage from images and video frames.
2. To comparatively benchmark these models on a common dataset split using standard metrics (accuracy, precision, recall, F1-score, mAP, IoU) to determine the most effective architecture for road damage detection.
3. To assess the feasibility of the best-performing lightweight model for real-time, edge-deployable automated road condition monitoring.

Importance / Motivation of the Proposed Project:

Road damage directly affects driver safety, vehicle wear, and public infrastructure budgets, yet most municipalities still rely on manual visual surveys that are slow, inconsistent between inspectors, and infeasible to run frequently across large road networks. An automated, camera-based deep learning pipeline can continuously monitor road condition at low cost using dashcams, drones, or municipal survey vehicles, enabling earlier detection of deterioration, data-driven prioritization of repair budgets, and a measurable comparison of which model family (classification, detection, segmentation, or attention-based) is most practical for real-world, real-time deployment.

Preliminary Reference List:

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[5] S. L. H. Lau, E. K. P. Chong, X. Yang, and X. Wang, "Automated Pavement Crack Segmentation Using U-Net-Based Convolutional Neural Network," IEEE Access, vol. 8, pp. 114892-114899, 2020.

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Work Plan:

Sl. No.	Component / Work Element / Milestone	Expected Completion (Day/Month)
1	Team formation, topic finalization, dataset (RDD2022) download & exploration	22 Aug 2026
2	Synopsis report writing & submission (Phase 1)	25 Aug 2026
3	Data preprocessing pipeline: augmentation, train/val/test split, video frame extraction	10 Sep 2026
4	Implement & train first 2 models (YOLOv8 detector, ResNet/MobileNet classifier) with preliminary results	20 Sep 2026
5	Interim report writing & submission (Phase 2)	25 Sep 2026